Homework

Modules 1–8 · ∼2.5 hours · Open the interactive tutorial while you work

Open the tutorial in your browser before starting. Every answer can be found by reading the modules and playing with the interactive widgets. Show your work where math is involved. Write your answer in the space provided. **Estimated time: 2 hours.**

Problem 1 — Signals

A patient's alpha rhythm has frequency 10 Hz and amplitude 30 μV.

(a) What is its period in milliseconds?

(b) How many full cycles fit in one second?

refer to: Module 1 — Signals

Problem 2 — Sampling

An EEG amplifier samples at 256 Hz.

(a) What is the highest signal frequency you can faithfully record?

(b) If a 200 Hz EMG burst contaminates the recording, at what frequency does it appear in the digitized signal?

refer to: Module 2 — Sampling

Problem 3 — Filtering

(a) European mains electrical noise sits at 50 Hz. What type of filter would you use to remove it?

(b) If you set the low-pass filter too aggressively (e.g., cut everything above 15 Hz), what important EEG feature might you miss, and why is that clinically dangerous?

refer to: Module 3 — Filtering

Problem 4 — FFT

You record a 4-second EEG segment sampled at 250 Hz.

(a) How many samples is that?

(b) What is the frequency resolution of an FFT of the whole segment?

(c) If you instead split the recording into 1-second windows and FFT each, what does the resolution become?

refer to: Module 4 — FFT

Problem 5 — EEG Frequency Bands

Match each brain state to its dominant EEG rhythm and state the approximate frequency band. (Hint: try the clinical-scenario presets in the Band Builder.)

Brain state	**Rhythm name**	**Frequency band (Hz)**
Deep sleep (N3)		
Drowsy		
Awake, eyes closed		
Focused mental task		
Active thinking / cognition		

refer to: Module 5 — EEG Basics

Problem 6 — The 10-20 Electrode Naming System

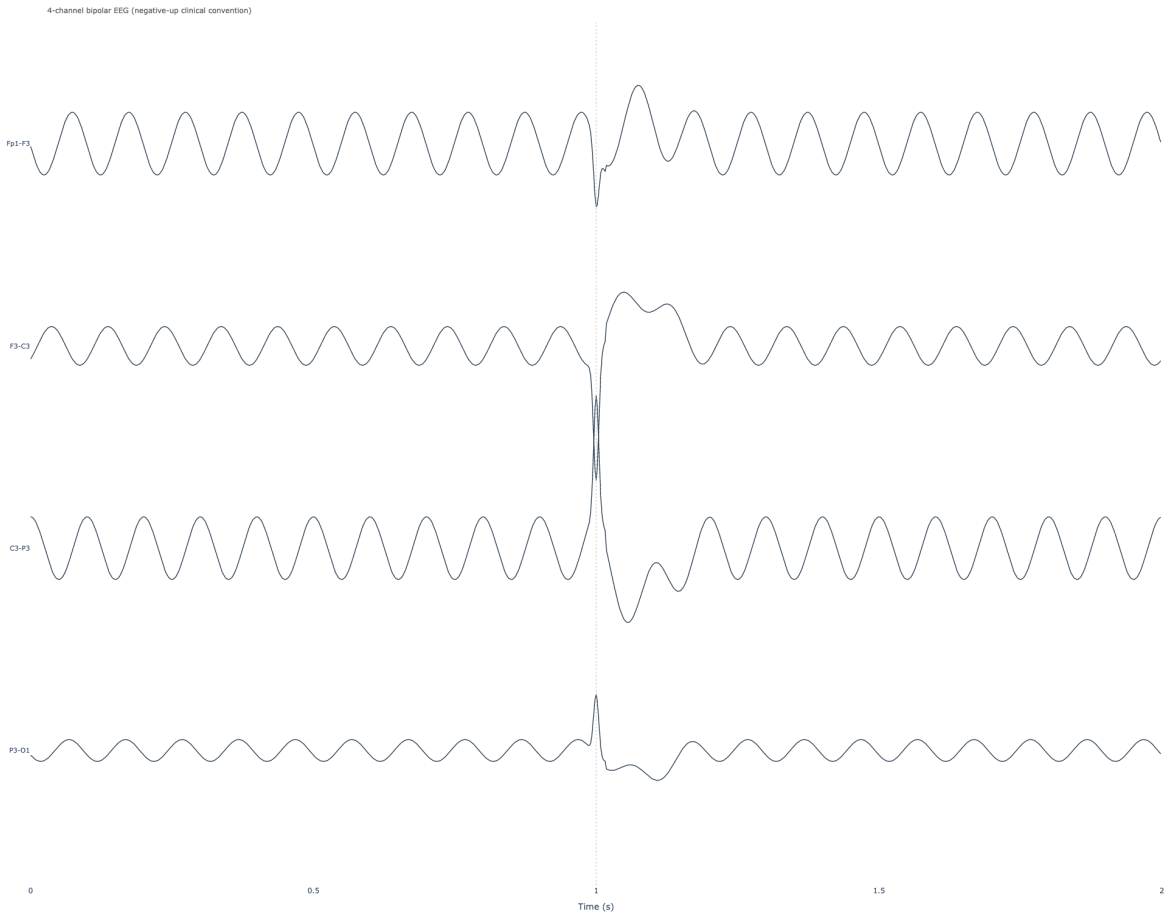
Each EEG electrode has a name with a letter and a number. For each electrode below, name the brain region it sits over and which side of the head it is on (left, right, or midline).

Electrode	Brain region	Side
Fp2		
T3		
Pz		
O1		

refer to: Module 5 — EEG Basics (10-20 System)

Problem 7 — Phase Reversal

Look at the four-channel bipolar EEG trace below. The display follows clinical convention (*negative voltage deflects upward*).



(a) Which electrode is closest to the spike source?

(b) Explain in 2 sentences how you can tell.

refer to: Module 5b — Montages

Problem 8 — The Double Banana

The longitudinal bipolar montage (the "double banana") is the clinical default for reading routine EEG.

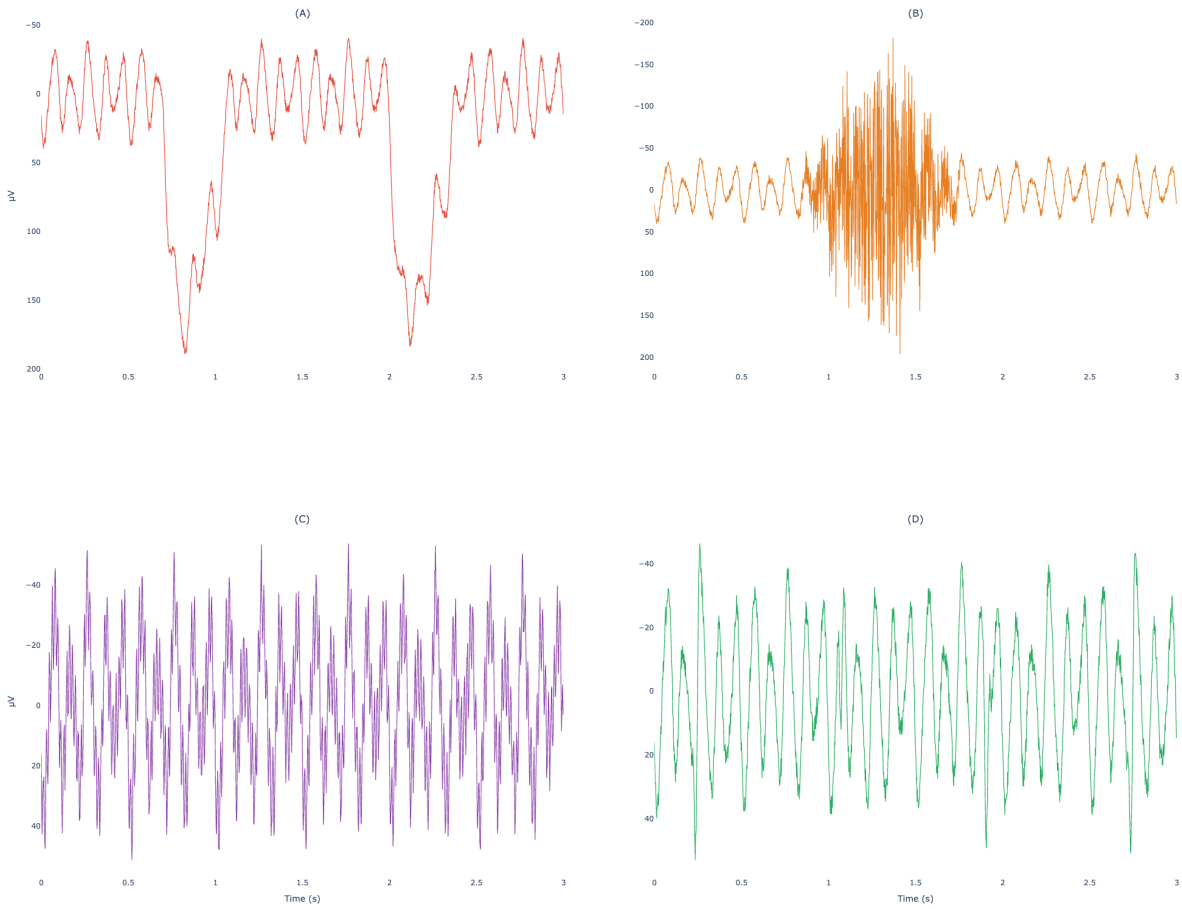
(a) List the four electrode chains of the double banana, naming the electrodes in each chain in order from front to back.

(b) All four chains run **front-to-back**. What kind of brain activity would this montage tend to underestimate, and why?

refer to: Module 5b — The Double Banana Montage

Problem 9 — Artifacts

The four panels below each show 3 seconds of EEG contaminated by a different artifact. Identify the artifact in each panel (eye blink, muscle, 60 Hz line noise, ECG). For **one** of the four, give a one-sentence removal strategy.



Panel	Artifact
A	
B	
C	
D	

Removal strategy (pick one panel):

refer to: Module 6 — Artifacts

Problem 10 — Spectrograms

Open the spectrogram widget in Module 7. Set the window length to **0.5 seconds**, observe the spectrogram, then change it to **2 seconds**.

(a) What improves and what gets worse when you go from 0.5 s to 2 s?

(b) For detecting the precise moment a seizure begins on EEG, which window would you choose? Why?

refer to: Module 7 — Spectrograms

Problem 11 — The Real Thing (Module 8)

Load the file `chb03_01.edf` (provided with this tutorial, or download from PhysioNet).

(a) How many channels are in the recording? What is the sampling rate? How long is the recording in minutes?

(b) What type of montage is this? (*Hint: look at the channel names.*)

(c) Scroll through the recording. Something abnormal happens. At approximately what time (in seconds) does it start? How long does it last?

(d) Compute a spectrogram for one channel around the abnormal event. Include the plot. Describe what you see in 1–2 sentences.

(e) Based on the clinical vignette in Module 8 and your analysis: what is happening to this patient? Answer in 2–3 sentences.

refer to: Module 8 — The Challenge